SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: Total Marks: 50
Annexure A

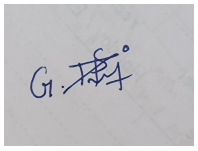
Scenario-Based

Code of Conduct:

I **G.DHANUSHA SREE(192325012)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**G.DHANUSHA SREE**



Annexure A Scenario-based

1.Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

1.Identify the most suitable algorithm for this scenario.

2.Justify your choice based on security strength and efficiency.

ANSWER:

1. Introduction

A banking application handles highly confidential information such as customer details, account information, transaction records, and payment data. If this information is exposed or altered, it can cause financial loss, fraud, and loss of customer trust. Therefore, a banking system needs a strong encryption algorithm that provides confidentiality while maintaining acceptable performance.

The algorithms given for comparison are DES (Data Encryption Standard), 3-DES (Triple DES), and Blowfish. All three are symmetric-key block ciphers, meaning that the same secret key, or related secret key material, is used to encrypt and decrypt data. However, they differ in key strength, design, performance, and present-day suitability.

2. Requirements of a Banking Application

The selected algorithm should meet the following requirements:

High confidentiality: transaction information must not be readable by unauthorized parties.

Strong resistance to attacks: the algorithm should withstand practical cryptanalytic attacks and brute-force attempts.

Moderate or high performance: encryption and decryption should not create unacceptable transaction delays.

Reliable implementation: the algorithm should be available in well-tested cryptographic libraries.

Support for secure protocols: it should work within a secure design that includes key management, authentication, and integrity protection.

3. Comparison of DES, 3-DES, and Blowfish

3.1 DES (Data Encryption Standard)

DES is a symmetric block cipher that processes data in 64-bit blocks and uses an effective key length of 56 bits. It was historically important, but its key is now too short to protect sensitive banking information against modern brute-force attacks. A sufficiently capable attacker can search the DES key space much more feasibly than was possible when DES was introduced. For this reason, DES should not be selected for a new banking system.

Advantages: DES is relatively simple and historically well studied. Disadvantages: its 56-bit key is inadequate for modern security, and its 64-bit block size is also unsuitable for protecting large volumes of data under a single key.

3.2 3-DES (Triple DES)

Triple DES applies the DES operation three times, commonly using an encrypt-decrypt-encrypt construction. It was introduced to strengthen DES without completely replacing its design. Depending on the keying option, its nominal key material can be larger than DES, and it is substantially more resistant to brute-force attacks than single DES.

However, 3-DES is slower than modern ciphers because it performs DES operations multiple times. It also retains a 64-bit block size, which creates practical limits when encrypting large

amounts of data. Triple DES is now considered legacy technology and is not recommended for new systems. Its use for new encryption has been disallowed or deprecated in major standards and protocols.

3.3 Blowfish

Blowfish is a symmetric block cipher designed by Bruce Schneier. It uses a 64-bit block size and supports variable-length keys from 32 bits up to 448 bits. It was designed to be efficient in software and can perform well in many general-purpose computing environments.

Blowfish has a much larger possible key length than DES, and it avoids the three-pass processing used by 3-DES. However, its 64-bit block size is a significant limitation for modern high-volume encryption, and it is not the preferred choice for a new banking application. A modern authenticated encryption algorithm such as AES-GCM or ChaCha20-Poly1305 is a better practical choice, although these are outside the three options in the question.

4. Comparison Table

Feature

Feature	DES	3-DES	Blowfish	Practical assessment
Cipher type	Symmetric block cipher	Symmetric block cipher	Symmetric block cipher	All use secret-key cryptography
Block size	64 bits	64 bits	64 bits	64-bit blocks are a legacy limitation
Key length	56-bit effective key	Varies by keying option	32–448 bits	Longer key size alone does not determine security
Performance	Fast in many legacy contexts	Slower; DES operation repeated	Often efficient in software	Performance depends on platform and implementation

If the answer must be restricted to DES, 3-DES, and Blowfish, Blowfish is the most reasonable classroom choice when considering key flexibility and software efficiency. DES must be rejected because its 56-bit key is too weak, while 3-DES offers stronger protection than DES but has poor performance compared with modern alternatives and is now legacy technology.

This is a relative answer among the listed options, not a recommendation to deploy Blowfish in a new banking product. For a real banking system, none of these three would be the preferred new-design choice. AES with a suitable authenticated encryption mode, such as AES-GCM, is generally a more appropriate standard option.

6. Justification Based on Security and Efficiency

Security strength

DES has an effective key length of only 56 bits and can be broken by exhaustive key search with sufficient resources. 3-DES improves on DES by applying the cipher multiple times, but its legacy status and 64-bit block size make it unsuitable for new banking encryption. Blowfish supports longer keys and has no known practical attack that simply reduces its security to DES's 56-bit brute-force level when used correctly; nevertheless, its 64-bit block size remains a concern for large data volumes.

Efficiency

Blowfish can be efficient for software encryption, whereas 3-DES requires several DES operations for each block and is usually slower. DES may be computationally light, but its weak key makes it unacceptable. Therefore, Blowfish is the most defensible choice among the three for a theoretical comparison emphasizing security and moderate performance, with the important qualification that a modern cipher should be used in a real banking system.

7. Additional Security Measures for Banking

An encryption algorithm alone does not make a banking system secure. The application should also include:

Authenticated encryption to protect confidentiality and detect data modification.

Secure key generation, storage, rotation, and access controls, ideally using a hardware security module where appropriate.

TLS for data transmitted between customers, applications, and banking servers.

Strong user authentication and transaction authorization.

Audit logs, monitoring, rate limiting, and secure software updates.

8. Conclusion

For a textbook comparison limited to DES, 3-DES, and Blowfish, Blowfish is the most suitable of the three because it supports longer keys and is generally more efficient than 3-DES. DES is not secure enough, and 3-DES is slow and outdated. For a real banking application, however, the recommended approach is a modern authenticated encryption standard such as AES-GCM, supported by robust key management and secure communication protocols.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

1.Analyze the constraints of IoT devices.

2.Recommend a suitable cryptographic approach and justify your answer.

ANSWER:

1. Introduction

The Internet of Things (IoT) consists of connected devices such as smart locks, temperature sensors, light controllers, security cameras, and home automation hubs. These devices exchange information over wired or wireless networks. Some information may be sensitive—for example, device credentials, control commands, occupancy information, or sensor

readings. Cryptography is therefore required to protect communication from eavesdropping, impersonation, and unauthorized modification.

The designer must choose between RSA and Elliptic Curve Cryptography (ECC). Both are public-key cryptographic approaches: they use a public key that can be shared and a corresponding private key that must be protected. Their computational and storage requirements differ significantly, which matters for low-power IoT devices.

2. Constraints of IoT Devices

2.1 Limited processing capacity

Many IoT devices use small microcontrollers rather than powerful processors. They may have limited clock speed and little hardware support for public-key mathematics. Cryptographic operations that take too long can delay device responses and affect the reliability of real-time actions, such as locking a door or triggering an alarm.

2.2 Limited memory and storage

IoT devices may have only a small amount of RAM and flash storage. Large public keys, certificates, cryptographic libraries, and intermediate calculations consume memory. Smaller keys and compact protocol messages can help reduce storage and communication overhead.

2.3 Restricted battery power

Many sensors and smart devices operate on batteries and may remain deployed for months or years. Cryptographic computations consume energy, as do radio transmissions. A solution that reduces computation and the number of bytes transmitted can extend battery life.

2.4 Network bandwidth and latency

IoT devices may communicate over low-bandwidth or high-latency networks. Large keys, certificates, and handshake messages increase the

amount of data transmitted and can slow the establishment of secure connections. Compact cryptographic mechanisms are therefore useful.

2.5 Physical exposure and long-term operation

Devices may be installed in homes, factories, or outdoor locations where attackers can physically access them. Private keys must be stored securely, firmware must be updated, and devices need protection against replay, impersonation, and unauthorized commands. Cryptographic algorithm selection must be accompanied by secure implementation and key management.

3. RSA Cryptosystem

RSA is a public-key cryptosystem whose security is based on the difficulty of factoring a large composite integer. It is widely known and has extensive implementation support. RSA can be used for digital signatures and, with appropriate padding, public-key encryption or key establishment.

Advantages of RSA

Well established and widely supported in cryptographic libraries and protocols.

Can provide digital signatures and public-key operations.

A small public exponent can make signature verification relatively fast.

Disadvantages of RSA for IoT

Requires relatively large keys for modern security, such as RSA-2048 or larger.

Large keys and signatures increase storage and network overhead.

Private-key operations can be computationally expensive on small processors.

Key generation and certificate processing may require substantial computation and memory.

A 1024-bit RSA key is not considered adequate for modern security. For new systems, RSA-2048 is a common minimum baseline, subject to the system's security policy and expected lifetime.

4. Elliptic Curve Cryptography (ECC)

ECC is a family of public-key cryptographic techniques based on mathematical operations on elliptic curves over finite fields. It can support key agreement and digital signatures. Common examples include ECDH for key agreement and ECDSA for digital signatures; modern systems may also use curve-based schemes such as X25519 and Ed25519 where supported.

Advantages of ECC

Provides strong security with substantially smaller keys than RSA.

Reduces key and signature sizes in many common schemes.

Can reduce storage and network communication overhead.

Often offers a favorable balance of performance and energy consumption on constrained devices.

Disadvantages and considerations

Implementation must use well-reviewed libraries and correctly selected curves.

Performance varies by curve, processor, and available hardware acceleration.

Poor random-number generation or private-key handling can undermine security.

ECC does not eliminate the need for authentication, certificate validation, or secure key storage.

5. RSA and ECC Comparison

Feature	RSA	ECC	IoT impact
Security for comparable strength	Requires larger keys	Uses much smaller keys	ECC reduces storage and message size
Processing	Private-key operations	Often efficient with	ECC is usually attractive for

	can be costly	suitable curves	constrained processors
Memory	Larger keys and signatures	Smaller keys and signatures	ECC typically needs less memory
Communication	More bytes for keys/signatures	Compact public keys/signatures	ECC reduces bandwidth use
Energy	May require more processing and transmission	Often lower overall cost	ECC commonly suits battery-powered devices
Ecosystem	Very broad support	Broad support, but scheme/curve support must be checked	Choose based on device and protocol compatibility

6. Recommended Cryptographic Approach

ECC is the recommended choice for the described low-power smart-home IoT devices, provided that the device and protocol support a well-reviewed ECC implementation. It provides comparable security with much smaller keys than RSA, reducing memory requirements, communication overhead, and often the energy needed for public-key operations.

6.1 Key agreement

Use an approved elliptic-curve key-agreement mechanism, such as ECDH or X25519 where appropriate, to establish shared secret material between the device and the trusted hub or server. The public keys exchanged over the network do not need to be secret, but they must be authenticated so that an attacker cannot substitute their own key.

6.2 Symmetric encryption for data

Public-key cryptography should generally not be used to encrypt every sensor reading or control message directly. Instead, use the established shared secret to derive symmetric session keys and encrypt communication using an authenticated encryption scheme such as AES-GCM or ChaCha20-Poly1305, depending on platform and protocol support. This approach is efficient for frequent data transfer.

6.3 Authentication and integrity

Devices should authenticate the hub or server and validate certificates or public keys according to the system's trust model. Authenticated encryption detects unauthorized message modification. Protocols should also protect against replay attacks using counters, nonces, sequence numbers, or other protocol-approved mechanisms.

6.4 Secure key storage and updates

Private keys and long-term secrets should be stored in protected memory or a secure element when available. Devices need a secure process for provisioning, revoking, and updating credentials. Firmware updates should be authenticated, and old cryptographic algorithms should be replaced when they no longer meet security requirements.

7. Justification for Choosing ECC

Security

ECC offers strong security with smaller public keys than RSA. For example, a 256-bit elliptic-curve key is commonly compared to RSA-3072 for roughly similar classical security strength. The exact comparison depends on the selected schemes and security estimates. A 160-bit ECC key, although sometimes used in older educational comparisons, is too weak for new systems.

Efficiency

Smaller keys reduce the amount of data that must be stored and transmitted. On constrained devices, this can reduce both memory use and radio energy. ECC operations are often efficient on IoT-class hardware, although the final result should be measured on the actual device because processor architecture and implementation quality can affect performance.

Scalability

A smart-home system may contain many sensors and controllers. Compact keys and handshake messages can reduce connection overhead across the device fleet. ECC is therefore a practical option for scalable

secure communication, provided key provisioning and lifecycle management are designed properly.

8. Conclusion

IoT devices operate under constraints such as limited processing capacity, memory, battery power, and network bandwidth. RSA is a well-known and widely supported public-key cryptosystem, but its relatively large keys and potentially costly private-key operations can be disadvantageous for small devices. ECC provides comparable modern security with much smaller keys and usually lower communication overhead, making it the preferred option for this scenario. A sound design uses ECC for authenticated key agreement or signatures and a symmetric authenticated encryption algorithm for the actual communication data. Secure implementation, key storage, certificate validation, and update mechanisms remain essential.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level4 (Exemplary)	Level 3 (Proficien t)	Level 2 (Developing)	Level1 (Beginnin g)
Understandin g of Scenario	20%	Demonstrat es clear and complete understandi ng of the scenario and context	Shows good understa ndi ng with minor gaps	Partial understandi ng; misses key aspects	Misinterpre ts or fails to understand the scenario
entification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifi es most key issues with minor omissio ns	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applie s conce pts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasona ble decisio ns with some justific ation	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	General ly clear with minor issues	Somewhat unclear or poorly structured	Disorganiz ed and difficult to understand